

For 2-dimensional vectors:

$$\text{Manhattan distance} = |x_1 - x_2| + |y_1 - y_2|$$

For 3-dimensional vectors:

$$\text{Manhattan distance} = |x_1 - x_2| + |y_1 - y_2| + |z_1 - z_2|$$

General formula for n-dimensional vectors:

$$\text{Manhattan distance} = |x_1 - x_2| + |y_1 - y_2| + \dots + |v_{n1} - v_{n2}|$$

Scalars, Vectors, Matrices & Tensors

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$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$\begin{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} & \begin{bmatrix} 3 & 2 \end{bmatrix} \\ \begin{bmatrix} 1 & 7 \end{bmatrix} & \begin{bmatrix} 5 & 4 \end{bmatrix} \end{bmatrix}$

Scalar

Vector

Matrix

Tensor